Bayesian Reasoning and Learning (IM0902)

Jurgen de Vries (852759710)

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Abstract

This study investigates the application of Bayesian networks in forensic investigations involving Y-chromosomal DNA (Y-STR) profile matching. A Bayesian network was constructed to combine information obtained from observed Y-STR matches, haplotype frequency, familial relationships, investigative leads and traditional evidence to assess the overall strength of evidence in a forensic case. The network was parameterized using findings from the forensic literature together with illustrative probabilities where empirical estimates were unavailable. Inference experiments demonstrated that the network behaved consistently with the intended forensic reasoning and correctly propagated evidence through the network. In addition, data generated from the network were used to investigate whether its structure could be recovered using both constraint-based and score-based structure learning algorithms. Although the learned structures differed from the original network, particularly in edge orientations, they achieved reasonable predictive performance. The results indicate that Bayesian networks provide a useful framework for integrating genetic forensic evidence and that effective classification is possible even when the original network structure cannot be perfectly recovered.

1 Introduction

This paper is inspired by *Dader onbekend* [1] and examines the application of Bayesian networks in forensic investigations involving Y-chromosomal DNA profile matching. The study focuses on incorporating information from genetic relatives with traditional forms of evidence to assess the probability of the overall strength of evidence in a case.

Y-chromosomal information held in Y-chromosome short tandem repeats (Y-STR), is present only in male DNA and, except for occasional mutations, is transmitted largely unchanged from father to son across generations. Unlike autosomal DNA, which is inherited from both parents, Y-STR follows the paternal lineage exclusively. This inheritance pattern makes it particularly suitable for identifying distant paternal relatives and tracing ancestral lineages, including relationships involving cousins, uncles, and nephews. [2, 3]. As discussed in *Dader onbekend* [1], Y-STR analysis and matching has been applied in several cold cases by testing on select populations to identify familial relationships with an unknown perpetrator [4].

Because Y-STR profiles are shared by paternal relatives and haplotype frequencies vary across populations, the evidential value of a Y-STR match must be interpreted cautiously. Incorrect interpretation may lead to false accusations and reputational damage [4].

The purpose of this study is not to model the process of Y-STR matching itself within a Bayesian network. Instead, the objective is to incorporate information derived from Y-chromosomal DNA matches as variables in a forensic Bayesian network to assess the overall strength of evidence in a case. The network considers not only an Y-STR matches and the degree familial relatedness, but also additional evidence, the rarity of the haplotype, investigative leads associated with the identified relatives, and other traditional evidence.

The Bayesian network should enable probabilistic reasoning about the strength of the overall evidence given the observed information. For example, if a positive Y-STR match has been found for a common haplotype, what is the probability that the combined evidence provides strong support for the hypothesis that a specific individual is the source of the DNA sample found on the crime scene?

In the second part of this study, the goal is to investigate whether the structure of the network can be relearned using data generated from the constructed Bayesian network, and whether the Bayesian networks can be used as a classifier. Rather than providing a comprehensive comparison of structure learning algorithms, this study explores a selection of learning methods and discusses the most notable findings.

2 Methods

2.1 Variables and connections

The selection of variables was guided by the forensic reasoning underlying Y-STR matching. The Y-chromosomal DNA is inherited through the paternal lineage and can therefore provide information about familial relationships between males [2, 3, 1]. Bieber et al. [2] explore the use of Y-chromosomal DNA for familial searching in offender databases. They demonstrate that additional information, such as geographic proximity, age, and surname, can be informative when evaluating potential familial relationship matches. For example, a DNA match in a state on the other side of the country than the crime was committed in, is less likely to represent a familial relationship. This same idea is further discussed by Andersen et al. [3] and Kennet et al. [4]. This motivated the inclusion of *Relationship* and *Investigative leads* variables. Where *Investigative leads* represents the new leads an inspector finds, which is influenced by the relationship of the found Y-STR match.

The direction of the dependency was chosen to represent the underlying causal process. Individuals who are closely related are more likely to share characteristics than unrelated individuals. Therefore, the variable *Relationship* was modeled as a parent of *Investigative leads*.

Andersen et al. [3] investigate the evidential value of Y-chromosomal DNA matches and discuss factors that influence the strength of such evidence. An important factor is the Y-chromosome haplotype, which refers to the specific combination of genetic markers observed on the Y-chromosome. Because Y-chromosomal DNA is inherited through the paternal lineage, individuals sharing a paternal ancestor are likely to share the same haplotype. However, some haplotypes occur much more frequently in the population than others. A match involving a rare haplotype provides stronger evidence of a familial relation than a match involving a common haplotype, which could be shared by many unrelated individuals. The importance of haplotype frequency in using Y-STR matches as evidence is confirmed by Kennet et al. [4]. For this reason, the *haplotype frequency* variable is adopted in the Bayesian network. When no Y-STR match can be found, the haplotype frequency alone would not provide evidence regarding the identity of the suspect. Therefore the *haplotype frequency* and *observed match* variables are combined into an intermediate *genetic evidence* variable representing the overall strength of genetic evidence.

Another aspect highlighted by Kennet et al. [4] is that a Y-STR match alone is rarely sufficient to identify an offender. Instead, such a match may lead investigators to relatives of the unknown offender, from which potential suspects can be identified. Additional evidence obtained from traditional investigation, such as witness statements, surveillance footage, or other forensic findings, must be considered together with the Y-STR match. This lead to adding an independent variable *other evidence* to the Bayesian network.

A final outcome variable *total evidence strength* is added to the Bayesian network to consider all types of evidence found: DNA match evidence, evidence from investigative leads, and other evidence together. This modeling choice is inspired by the use of Bayesian networks in forensic science where various forms of evidence are integrated in a Bayesian network to reason about their joint implications under uncertainty as discussed by Biederman et al. [5]. Rather than independently evaluating each source, the network provides a probabilistic assessment of the overall strength of the combined information, see figure 1.

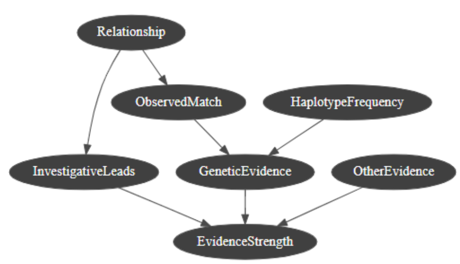


Figure 1: Resulting Bayesian network.

2.2 Parameters and priors

The variables in the Bayesian network are represented by a limited number of states to keep the network interpretable while still allowing it to represent a wide range of forensic scenarios:

1. The *relationship* variable distinguishes between {close relation, distant relation, no relation}.
2. The *haplotype frequency* variable is represented by the states {rare, common}.
3. The *observed match* variable is represented by the states {match, no match}.
4. The *investigative leads* variable is represented by the states {useful, not useful}.
5. The *genetic evidence* is represented by the states {strong, weak}.
6. The *other evidence* is represented by the states {supportive, neutral, not supportive}.
7. The *overall evidence* is represented by the states {strong, weak}.

Both Andersen et al. [3] en Kennet et al. [4] discuss that finding appropriate probabilities for variables involved in Y-STR matching is highly dependent on the population and the investigation context. For example, searching for a familial match in the population of an entire national population differs significantly from searching in a national offenders database or a population in a specific geographic region. However, the literature does provide guidance on the relative ordering of propbabilites. Close paternal relationships are less common than distant relationships, which in turn are less common then unrelated individuals. Similarly, useful investigative leads are more likely to be found when a close relative has been identified than when no relationship is present, and an observed Y-STR match is more likely for related individuals than for unrelated individuals.

Consequently, precise prior and conditional probabilities for the *relationship*, *haplotype frequency*, *observed match*, and *investigative leads* variables could not be obtained directly from the literature. Instead, illustrative probabilities were chosen that reflect the relative ordering of probabilities, while keeping in mind the task of relearning the structure.

The variable *genetic evidence* is determined by the *haplotype frequency* and *observed match* variables. Given no observed Y-STR match, the probability of strong genetic evidence is low regardless of the haplotype frequency. When a Y-STR match is observed however, the evidential strength depends on the rarity of the matching haplotype. A match involving a common haplotype provides Intermediate genetic evidence, whereas a match involving a rare haplotype provides strong genetic evidence due to its greater discriminating power.To keep the network interpretable, genetic evidence was modeled as a binary variable. Intermediate levels of genetic evidential strength are implicitly represented through the conditional probability table rather than introducing an additional state.

The type and strength of evidence in the *other evidence* variable is highly case dependent, and no generally applicable prior distributions could be obtained from the literature. Therefore, the prior probabilities were chosen to represent a neutral starting point for the given states, allowing the influence of case specific evidence to be explored during inference.

For the *Overall Evidence* representing the combined evidential value of all available information in the investigation, probabilities are chosen that reflect the available evidence from different sources.

The resulting prior and conditional probabilities used in the Bayesian network are presented in the conditional probability tables found in Appendix A.

2.3 Relearning structure

A comparison was made between reconstructing the network structure from the conditional independencies implied by the original Bayesian network and learning the structure from generated data using a constraint-based algorithm and a score-based algorithm.

First, the conditional independencies implied by the original Bayesian network were used as input for the PC-algorithm. The PC-algorithm is a constraint-based algorithm that utilizes conditional independence tests to construct the model [6]. Starting from a fully connected undirected graph, edges between conditionally independent variables were removed. Subsequently unshielded colliders were oriented and Meek's orientation rules were applied to infer additional edge directions. This procedure yields a partially directed acyclic graph (PDAG) implied by the original Bayesian network.

Second, datasets of varying sizes (100, 500, 1000 samples) were generated from the original Bayesian network. These datasets were used to learn network structures using both the MIIC algorithm [7], representing a constraint-based approach, similar to the PC-algorithm, and the Greedy Hill Climbing algorithm (GHC), representing a search-and-score-based approach. GHC is a search algorithm that starts with an initial DAG and repeatedly adds, deletes, or reverses edges, and evaluates resulting DAGs with a chosen score (e.g. BIC). For both methods the pyagrum implementation was used.

The learned structures were compared with the original Bayesian network and the PDAG. Both methods were evaluated using the Structural Hamming Distance (SHD), which counts the number of missing, added, and incorrectly oriented edges. Lower SHD scores indicate greater structural similarity.

3 Results

3.1 Inference

To see the workings of the constructed Bayesian network some configurations with different combinations of observations were tested

3.1.1 No observations

Without any observations, the posterior probability of *EvidenceStrength* is expected to favor the *weak* state. This follows from the prior probabilities in the network. Most individuals in the population are assumed to be unrelated, making observed Y-STR matches unlikely. In addition, common haplotypes occur more frequently than rare haplotypes, reducing the genetic evidential value of a potential match. Similarly, investigative leads are unlikely in the absence of a familial relationship, while the prior distribution of the *OtherEvidence* variable provides only limited support. Together, these factors result in a high prior probability of weak overall evidence (2).

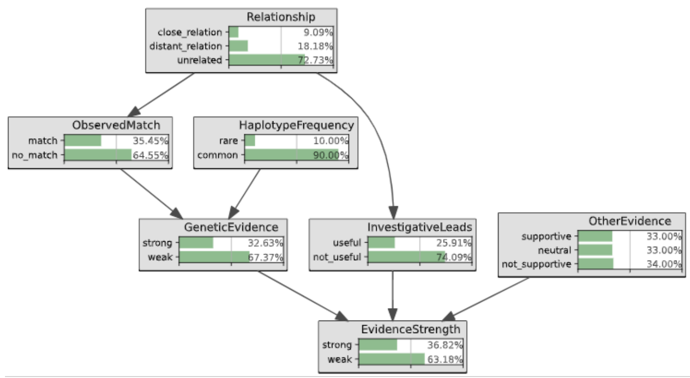


Figure 2: Inference with no observations

3.1.2 Observing a genetic match

Observing a Y-STR match increases the probability of strong genetic evidence and consequently increases the probability of overall evidence.

The increase in genetic evidence is larger than could be expected from an observed match alone, this is because the posterior probability of a close or distant familial relationship is increased as well through the dependency between these variables. This updated belief propagates through the network, increasing both the probability of usefull investigative leads and the overall evidence strength.

An additional observation is that the collider structures correctly block the propagation of information towards the *HaplotypeFrequency* and *OtherEvidence* variables (3).

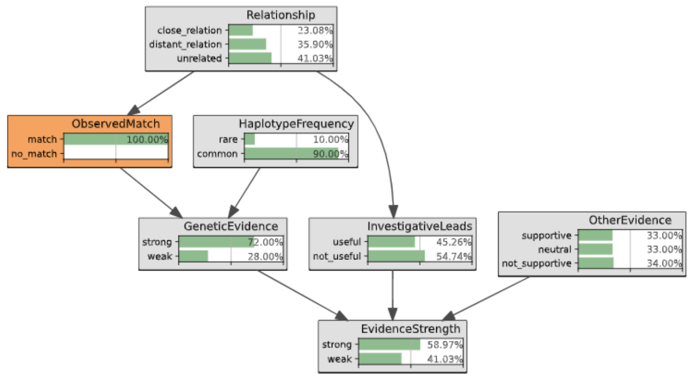


Figure 3: Inference with *ObservedMatch* variable observed.

3.1.3 Observing a match with a rare haplotype

Given the isolated position of the Haplotype behind a collider it can only influence the genetic evidence and overall evidence strength, and its influence doesn't propagate to other variables. Overall evidence strength in turn blocks the propagation to other variables due to its collider function as well. 4

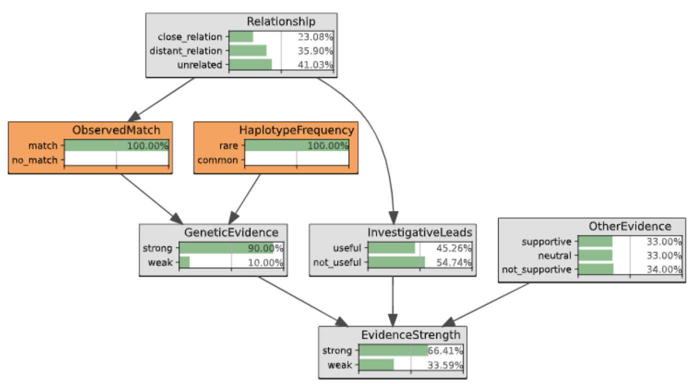


Figure 4: Inference with *ObservedMatch* and *HaplotypeFrequency* variables observed.

More inference examples can be found in Appendix B.

3.2 Learning the structure from data

Applying the PC algorithm to the conditional independencies implied by the original Bayesian network resulted in a PDAG in which all unshielded colliders were oriented, while two edges remained undirected (Appendix C). This PDAG represents three Markov equivalent DAGs, including the original network. Consequently, a learned DAG may differ by one edge orientations (SHD = 1) while still belonging to the same equivalence class.

Using the obtained independencies from the original network in combination with the PC-algorithm, led to a PDAG where the collider structures can be filled in (Appendix C). But leaves 2 edges that can not be oriented. This leads to 3 Markov Equivalent possible DAGs including the original network, resulting in a SHD score of 1 or 0.

The learning results in table 1 show that recovering the original DAG from observational data alone is difficult. The learned structures for each sample size and learning method combination are shown in Appendix D. In nearly all learned structures, at least one edge is oriented differently from the original DAG. This can be explained by the fact that several dependencies in the network can be interpreted in both directions. For example a close relationship increases the probability of observing a Y-STR match, while observing a Y-STR match also increases the posterior probability of a close relationship. This phenomenom was also observed during the inference experiments. Only one learned structure introduced an additional edge that was not present in the original network.

Despite these consistent orientation differences, several parts of the network were recovered consistently. The connections between *Relationship* and *ObservedMatch*, *Relationship* and *InvestigativeLeads*, and *ObservedMatch* and *GeneticEvidence* were present in all learned structures, although their directions were frequently reversed. Similarly, the connections between *EvidenceStrength* and its parent variables *InvestigativeLeads*, *OtherEvidence*, and *GeneticEvidence* were consistently recovered, also differently oriented most of the times.

A final observation concerns the *HaplotypeFrequency* and *OtherEvidence* variables. As shown in the inference experiments, both variables have a relatively isolated position in the network and exert only a limited influence on the remaining variables. This is reflected in several of the learned structures, particularly those learned from the smaller datasets, where one or both variables become completely disconnected from the rest of the network (Figure 5).

(a) MIIC (100 samples)

(b) Greedy Hill Climbing (100 samples)

Figure 5: Network structures learned from a dataset of 100 samples. In both networks, the *OtherEvidence* and *HaplotypeFrequency* variables are disconnected from the main network.

Table 1: Structural Hamming Distance (SHD) of the learned network structures.

Algorithm	Samples	Missing	Added	Mis oriented	Total
MIIC	100	2	0	1	3
MIIC	500	0	0	4	4
MIIC	1000	1	0	3	4
GHC	100	3	0	2	5
GHC	500	0	0	2	2
GHC	1000	0	1	3	4

3.3 Bayesian network as a classifier

To evaluate whether the learned Bayesian networks can be used as classifiers, the original network, the learned GHC network, and a Naive Bayes classifier were evaluated on a test set of 100 samples. The original Bayesian network achieved a ROC AUC of 0.8058 and a PR AUC of 0.9078. Despite structural differences from the original network, the GHC network learned from only 100 samples achieved a reasonable performance, with a ROC AUC of 0.7528 and a PR AUC of 0.8758. The Naive Bayes classifier obtained a ROC AUC of 0.70 and a PR AUC of 0.7999.

Although the learned network differs considerably from the original structure, its predictive performance remains similar. This indicates that accurately recovering every edge of the original Bayesian network is not necessary to obtain good classification performance. Capturing the most important probabilistic dependencies appears sufficient to make reliable predictions. The Naive Bayes classifier still performs reasonably well, although it is outperformed by the Bayesian network models.

6, 7, 8a, and 8b

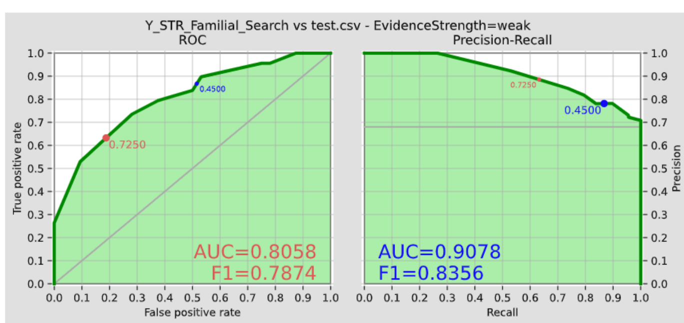


Figure 6: Constructed Bayesian network performance on 100 test samples.

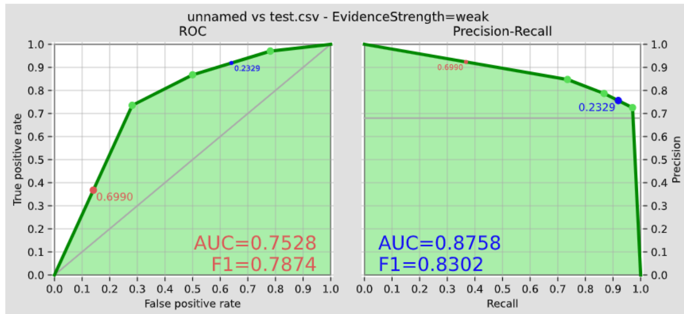
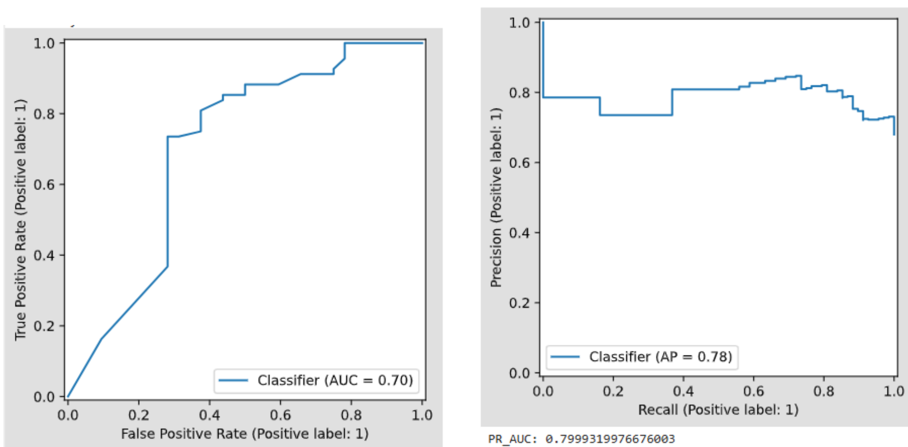


Figure 7: GHC relearned structure from 100 samples. Performance 100 test samples.



(a) ROC curve

(b) Precision–Recall curve

Figure 8: Performance of the Naive Bayes classifier on the test set of 100 samples.

4 Conclusion and discussion

The objective of this study was to investigate whether Y-STR matching can be incorporated into a Bayesian network to reason about the combined strength of forensic evidence, and whether such a network can be relearned from generated data. The constructed Bayesian network behaved as expected during inference. Observations propagated through the network in a logical manner, while the collider structures prevented information from flowing to variables that should remain independent of each other. This demonstrates that the chosen network structure is suitable for combining genetic evidence with traditional forms of evidence.

Relearning the the structure from observational data proved to be more challenging. Although many of the relevant dependencies were consistently recovered, edge directions frequently differed from those in the original network. This can largely be explained by the fact that several dependencies are difficult to orient uniquely from observational data alone, resulting in structures that represent a similar set of of conditional independencies. The isolated position of variables such as *HaplotypeFrequency* and *OtherEvidence* also made these variables more difficult to recover, particularly for smaller datasets.

Despite these structural differences, the learned Bayesian networks achieved predictive performance comparable to that of the original network. Even the Naive Bayes classifier, despite its strong conditional independence assumptions, produced reasonable ROC AUC and Precision-recall scores. This suggests that accurate prediction does not necessarily require perfect recovery of the underlying network structure, provided that the most influential probabilistic relationships are captured.

Several simplifications were made to keep the network interpretable and manageable. One factor that could be accounted for is the DNA sample quality which influences the genetic evidence. Future work could extend the network with this real world factor and by estimating the conditional probabilities from empirical forensic data or expert judgment.

References

[1] Sofie Claerhout. *Dader onbekend*. Uitgeverij Lannoo, 2022.

[2] Frederick R Bieber, Charles H Brenner, and David Lazer. Finding criminals through dna of their relatives, 2006.

[3] Mikkel M Andersen and David J Balding. How convincing is a matching y-chromosome profile? *PLoS Genetics*, 13(11):e1007028, 2017.

[4] Debbie Kennett. Using genetic genealogy databases in missing persons cases and to develop suspect leads in violent crimes. *Forensic science international*, 301:107–117, 2019.

[5] A Biedermann and F Taroni. Bayesian networks for evaluating forensic dna profiling evidence: a review and guide to literature. *Forensic Science International: Genetics*, 6(2):147–157, 2012.

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[7] Louis Verny, Nadir Sella, Séverine Affeldt, Param Priya Singh, and Hervé Isambert. Learning causal networks with latent variables from multivariate information in genomic data. *PLoS computational biology*, 13(10):e1005662, 2017.

A Conditional Probability Tables

Relationship		
close_relation	distant_relation	unrelated
0.1000	0.2000	0.8000

(a) Relationship

HaplotypeFrequency	
rare	common
0.1000	0.9000

(b) Haplotype Frequency

	ObservedMatch	
Relationship	match	no_match
close_relation	0.9000	0.1000
distant_relation	0.7000	0.3000
unrelated	0.2000	0.8000

(c) Observed Match

		GeneticEvidence	
ObservedMatch	HaplotypeFrequency	strong	weak
match	rare	0.9000	0.1000
	common	0.7000	0.3000
no_match	rare	0.2000	0.8000
	common	0.1000	0.9000

(d) Genetic Evidence

	InvestigativeLeads	
Relationship	useful	not_useful
close_relation	0.8500	0.1500
distant_relation	0.6000	0.4000
unrelated	0.1000	0.9000

(e) Investigative Leads

OtherEvidence		
supportive	neutral	not_supportive
0.3300	0.3300	0.3400

(f) Other Evidence

Figure 9: Conditional probability tables for the Bayesian network variables.

			EvidenceStrength	
OtherEvidence	InvestigativeLeads	GeneticEvidence	strong	weak
supportive	useful	strong	0.9500	0.0500
		weak	0.6000	0.4000
	not_useful	strong	0.8500	0.1500
		weak	0.2500	0.7500
neutral	useful	strong	0.8000	0.2000
		weak	0.5000	0.5000
	not_useful	strong	0.6000	0.4000
		weak	0.1500	0.8500
not_supportive	useful	strong	0.7000	0.3000
		weak	0.3000	0.7000
	not_useful	strong	0.4000	0.6000
		weak	0.0500	0.9500

Figure 10: Conditional probability table for the Overall Evidence Strength variable.

B More inference examples

B.1 Genetic evidence is all you need?

The inference example in figure 11 demonstrates that genetic evidence alone is insufficient to conclude that the overall evidence is strong. Instead the Bayesian network requires support from additional sources of evidence before assigning a high probability to strong overall evidence.

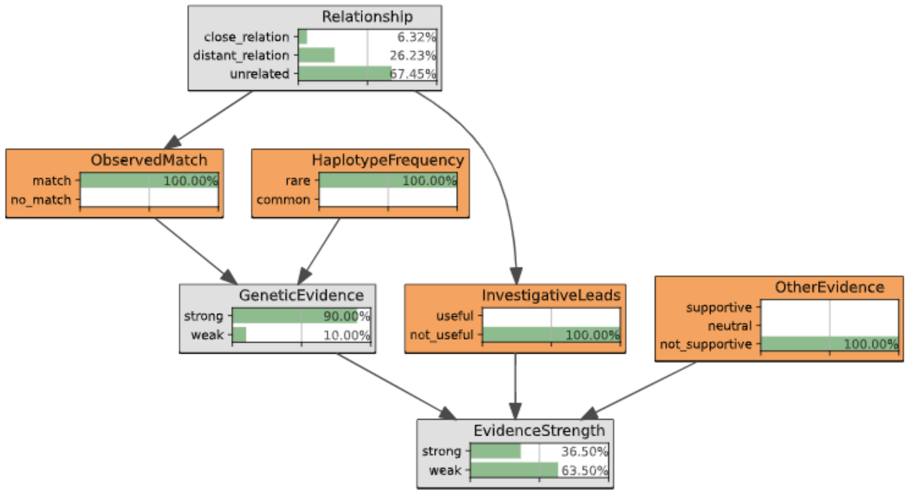

Figure 11: Genetic evidence is all you need? inference example.

This same phenomenom is observed with a single other source of evidence, see figures 12 and 13. While combining all sources of evidence results in a high probability of strong overall evidence see figure 14.

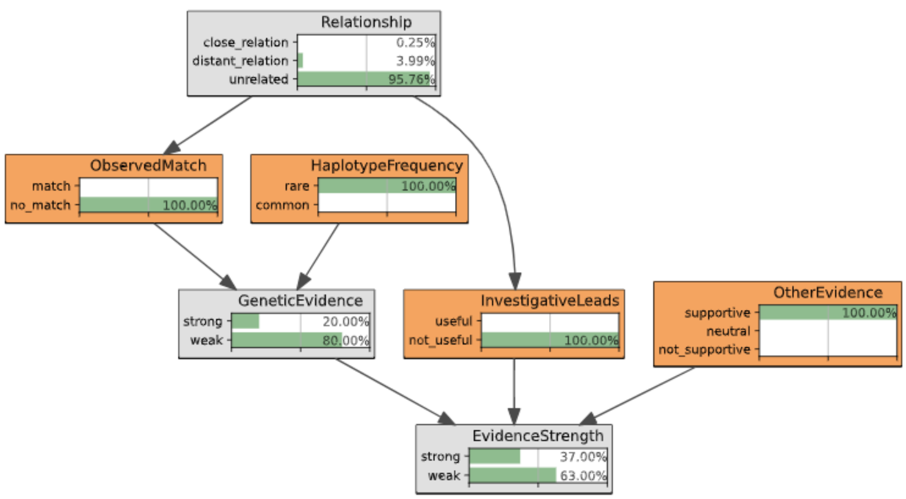

Figure 12: Observing only supportive *OtherEvidence*.

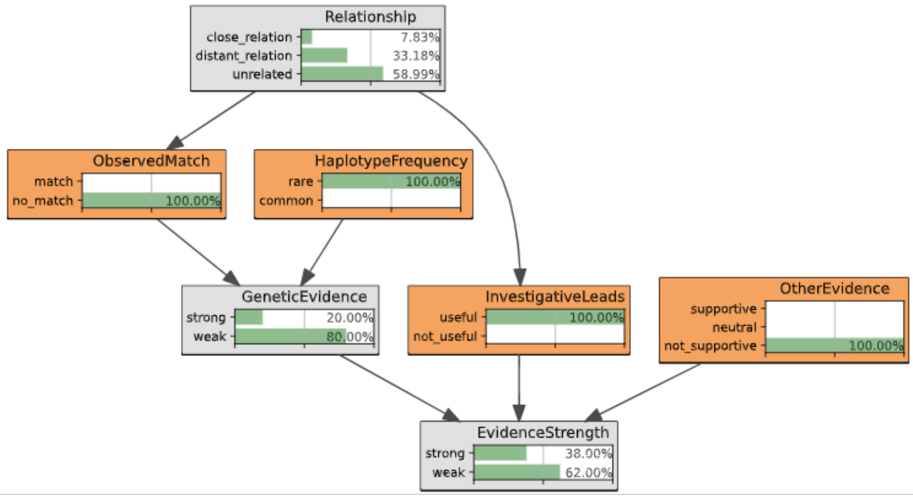

Figure 13: Observing only useful *InvestigativeLeads*.

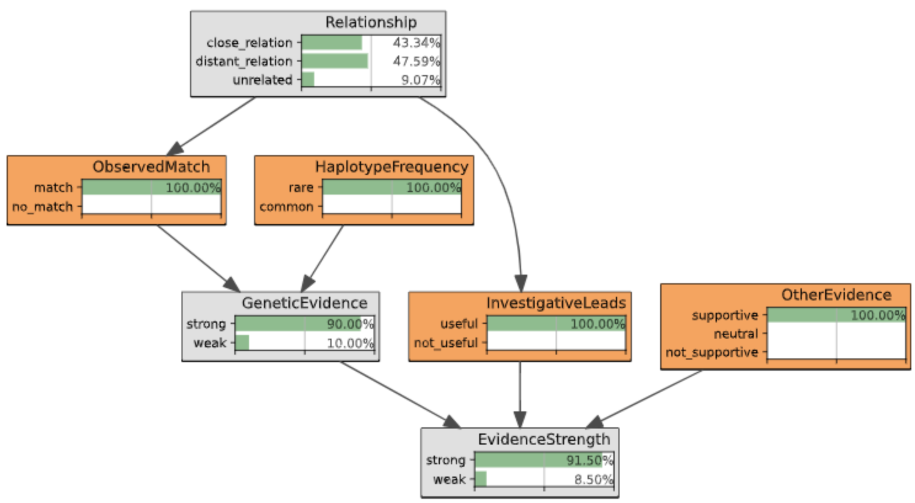

Figure 14: Combination of all types of evidence results in high probability of strong overall evidence.

One more interesting observation is that knowing the variables *ObservedMatch* and *InvestigativeLeads*, the actual relationship becomes obsolete for the overall evidence strength. Because all paths have been blocked, see figures 14 and 15.

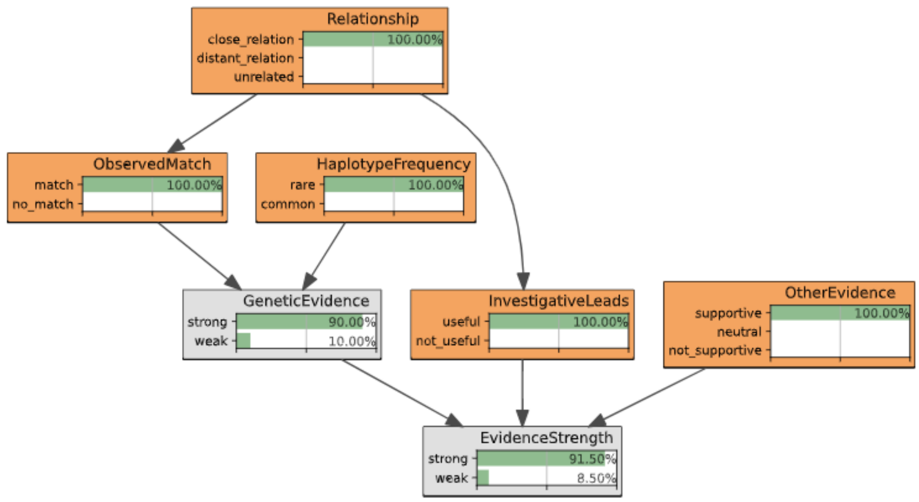

Figure 15: Combination of all types of evidence, and *Relationship* included, does not change outcome of *EvidenceStrength*.

C Conditional independencies and reconstructing the network using PC algorithm

The independencies extracted from the original Bayesian network using *pyagrum* give the necessary information to apply the PC-algorithm, and recover the PDAG that could lead to the original Bayesian network or a Markov equivalent version of it. Note that only dependencies with a conditioning set of at most 2 variables are included, these independencies contain enough information to remove all redundant connections.

$Relationship \perp HaplotypeFrequency|\{\}$
$Relationship \perp OtherEvidence|\{\}$
$Relationship \perp GeneticEvidence|\{ObservedMatch\}$
$Relationship \perp EvidenceStrength|\{ObservedMatch, InvestigativeLeads\}$
$Relationship \perp EvidenceStrength|\{InvestigativeLeads, GeneticEvidence\}$
$ObservedMatch \perp InvestigativeLeads|\{Relationship\}$
$ObservedMatch \perp HaplotypeFrequency|\{\}$
$ObservedMatch \perp OtherEvidence|\{\}$
$ObservedMatch \perp EvidenceStrength|\{Relationship,' GeneticEvidence\}$
$ObservedMatch \perp EvidenceStrength|\{InvestigativeLeads, GeneticEvidence\}$
$InvestigativeLeads \perp HaplotypeFrequency|\{\}$
$InvestigativeLeads \perp OtherEvidence|\{\}$
$InvestigativeLeads \perp GeneticEvidence|\{Relationship\}$
$InvestigativeLeads \perp GeneticEvidence|\{ObservedMatch\}$
$HaplotypeFrequency \perp OtherEvidence|\{\}$

$HaplotypeFrequency \perp EvidenceStrength | \{Relationship, GeneticEvidence'\}$
$HaplotypeFrequency \perp EvidenceStrength | \{ObservedMatch, GeneticEvidence\}$
$HaplotypeFrequency \perp EvidenceStrength | \{InvestigativeLeads, GeneticEvidence\}$
$OtherEvidence \perp GeneticEvidence | \{\}$

Starting from a fully connected undirected graph in figure 16.
To an undirected graph with independent connections removed in figure 17.

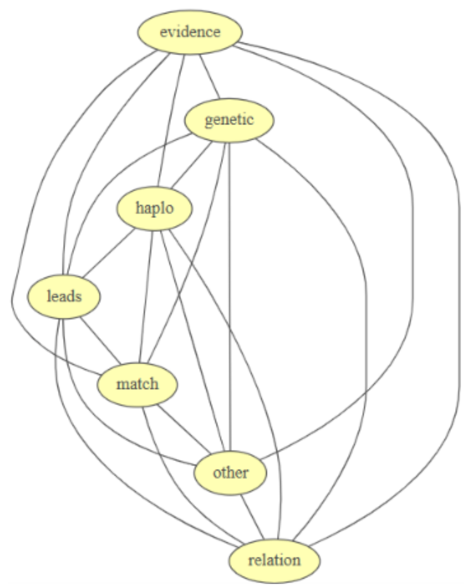

Figure 16: Fully connected

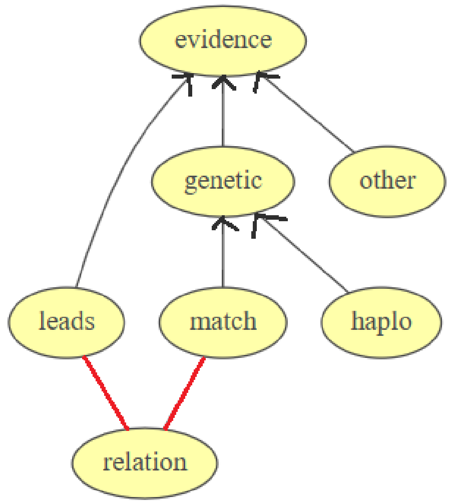

Figure 17: Independent connections removed and partially directed

Dependencies in the learned datasets differ from the dependencies implied by the original Bayesian network, making it harder to learn the structure from generated data alone.

C.1 Independencies found in dataset of 100 samples

$Relationship \perp ObservedMatch|\{InvestigativeLeads, HaplotypeFrequency, OtherEvidence\}$
$Relationship \perp ObservedMatch|\{InvestigativeLeads, HaplotypeFrequency, GeneticEvidence\}$
$Relationship \perp ObservedMatch|\{InvestigativeLeads, OtherEvidence, GeneticEvidence\}$
$Relationship \perp ObservedMatch|\{HaplotypeFrequency, OtherEvidence, GeneticEvidence\}$
$Relationship \perp ObservedMatch|\{HaplotypeFrequency, OtherEvidence, EvidenceStrength\}$
$Relationship \perp InvestigativeLeads|\{ObservedMatch, OtherEvidence, EvidenceStrength\}$
$Relationship \perp HaplotypeFrequency|\{\}$
$Relationship \perp OtherEvidence|\{\}$
$Relationship \perp GeneticEvidence|\{ObservedMatch\}$
$Relationship \perp GeneticEvidence|\{InvestigativeLeads\}$
$Relationship \perp GeneticEvidence|\{HaplotypeFrequency\}$
$Relationship \perp GeneticEvidence|\{EvidenceStrength\}$
$Relationship \perp EvidenceStrength|\{ObservedMatch\}$
$Relationship \perp EvidenceStrength|\{InvestigativeLeads\}$
$Relationship \perp EvidenceStrength|\{HaplotypeFrequency\}$
$Relationship \perp EvidenceStrength|\{GeneticEvidence\}$
$ObservedMatch \perp InvestigativeLeads|\{Relationship\}$
$ObservedMatch \perp HaplotypeFrequency|\{\}$
$ObservedMatch \perp OtherEvidence|\{\}$
$ObservedMatch \perp GeneticEvidence|\{Relationship, OtherEvidence, EvidenceStrength\}$
$ObservedMatch \perp EvidenceStrength|\{Relationship\}$

$ObservedMatch \perp EvidenceStrength|\{InvestigativeLeads\}$
$ObservedMatch \perp EvidenceStrength|\{GeneticEvidence\}$
$InvestigativeLeads \perp HaplotypeFrequency|\{\}$
$InvestigativeLeads \perp OtherEvidence|\{\}$
$InvestigativeLeads \perp GeneticEvidence|\{Relationship\}$
$InvestigativeLeads \perp GeneticEvidence|\{ObservedMatch\}$
$InvestigativeLeads \perp GeneticEvidence|\{HaplotypeFrequency\}$
$InvestigativeLeads \perp GeneticEvidence|\{OtherEvidence\}$
$InvestigativeLeads \perp GeneticEvidence|\{EvidenceStrength\}$
$InvestigativeLeads \perp EvidenceStrength|\{Relationship\}$
$HaplotypeFrequency \perp OtherEvidence|\{\}$
$HaplotypeFrequency \perp GeneticEvidence|\{\}$
$HaplotypeFrequency \perp EvidenceStrength|\{\}$
$OtherEvidence \perp GeneticEvidence|\{\}$
$OtherEvidence \perp EvidenceStrength|\{\}$
$GeneticEvidence \perp EvidenceStrength|\{Relationship, ObservedMatch, InvestigativeLeads\}$
$GeneticEvidence \perp EvidenceStrength|\{Relationship, ObservedMatch, HaplotypeFrequency\}$
$GeneticEvidence \perp EvidenceStrength|\{Relationship, ObservedMatch, OtherEvidence\}$
$GeneticEvidence \perp EvidenceStrength|\{Relationship, InvestigativeLeads, HaplotypeFrequency\}$
$GeneticEvidence \perp EvidenceStrength|\{ObservedMatch, HaplotypeFrequency, OtherEvidence\}$

C.2 Independencies found in dataset of 500 samples

$Relationship \perp HaplotypeFrequency|\{\}$
$Relationship \perp OtherEvidence|\{\}$
$Relationship \perp GeneticEvidence|\{ObservedMatch\}$
$Relationship \perp EvidenceStrength|\{ObservedMatch\}$
$ObservedMatch \perp InvestigativeLeads|\{Relationship\}$
$ObservedMatch \perp HaplotypeFrequency|\{\}$
$ObservedMatch \perp OtherEvidence|\{\}$
$ObservedMatch \perp EvidenceStrength|\{GeneticEvidence\}$
$InvestigativeLeads \perp HaplotypeFrequency|\{\}$
$InvestigativeLeads \perp OtherEvidence|\{\}$
$InvestigativeLeads \perp GeneticEvidence|\{Relationship\}$
$InvestigativeLeads \perp GeneticEvidence|\{ObservedMatch\}$
$InvestigativeLeads \perp GeneticEvidence|\{EvidenceStrength\}$
$InvestigativeLeads \perp EvidenceStrength|\{Relationship, ObservedMatch, HaplotypeFrequency\}$
$InvestigativeLeads \perp EvidenceStrength|\{Relationship, HaplotypeFrequency, OtherEvidence\}$
$HaplotypeFrequency \perp OtherEvidence|\{\}$
$HaplotypeFrequency \perp GeneticEvidence|\{Relationship, OtherEvidence\}$
$HaplotypeFrequency \perp GeneticEvidence|\{Relationship, EvidenceStrength\}$
$HaplotypeFrequency \perp GeneticEvidence|\{OtherEvidence, EvidenceStrength\}$
$HaplotypeFrequency \perp EvidenceStrength|\{\}$
$OtherEvidence \perp GeneticEvidence|\{\}$

C.3 Independencies found in dataset of 1000 samples

$Relationship \perp HaplotypeFrequency|\{\}$
$Relationship \perp OtherEvidence|\{\}$
$Relationship \perp GeneticEvidence|\{ObservedMatch\}$
$Relationship \perp EvidenceStrength|\{InvestigativeLeads\}$
$ObservedMatch \perp InvestigativeLeads|\{Relationship\}$
$ObservedMatch \perp HaplotypeFrequency|\{\}$
$ObservedMatch \perp OtherEvidence|\{\}$
$ObservedMatch \perp EvidenceStrength|\{GeneticEvidence\}$
$InvestigativeLeads \perp HaplotypeFrequency|\{\}$
$InvestigativeLeads \perp OtherEvidence|\{\}$
$InvestigativeLeads \perp GeneticEvidence|\{Relationship\}$
$InvestigativeLeads \perp GeneticEvidence|\{ObservedMatch\}$
$HaplotypeFrequency \perp OtherEvidence|\{\}$
$HaplotypeFrequency \perp GeneticEvidence|\{InvestigativeLeads\}$
$HaplotypeFrequency \perp GeneticEvidence|\{OtherEvidence\}$
$HaplotypeFrequency \perp GeneticEvidence|\{EvidenceStrength\}$
$HaplotypeFrequency \perp EvidenceStrength|\{Relationship\}$
$HaplotypeFrequency \perp EvidenceStrength|\{InvestigativeLeads\}$
$HaplotypeFrequency \perp EvidenceStrength|\{OtherEvidence\}$
$HaplotypeFrequency \perp EvidenceStrength|\{GeneticEvidence\}$
$OtherEvidence \perp GeneticEvidence|\{\}$

D Learned structures, essential graphs, and graph differences

Below the differences in the original Bayesian network (left) and the learned structures (right) are displayed.

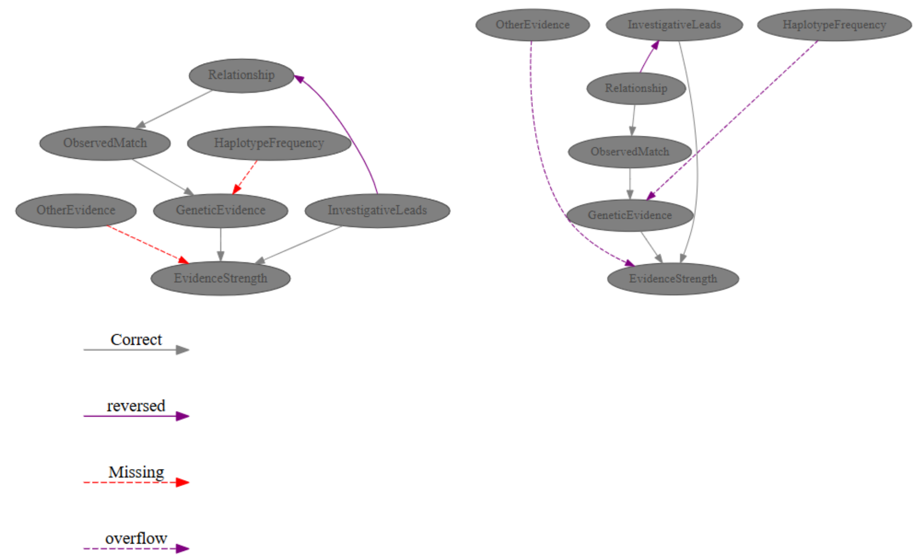

Figure 18: MIIC-algorithm, learned structure from 100 samples.

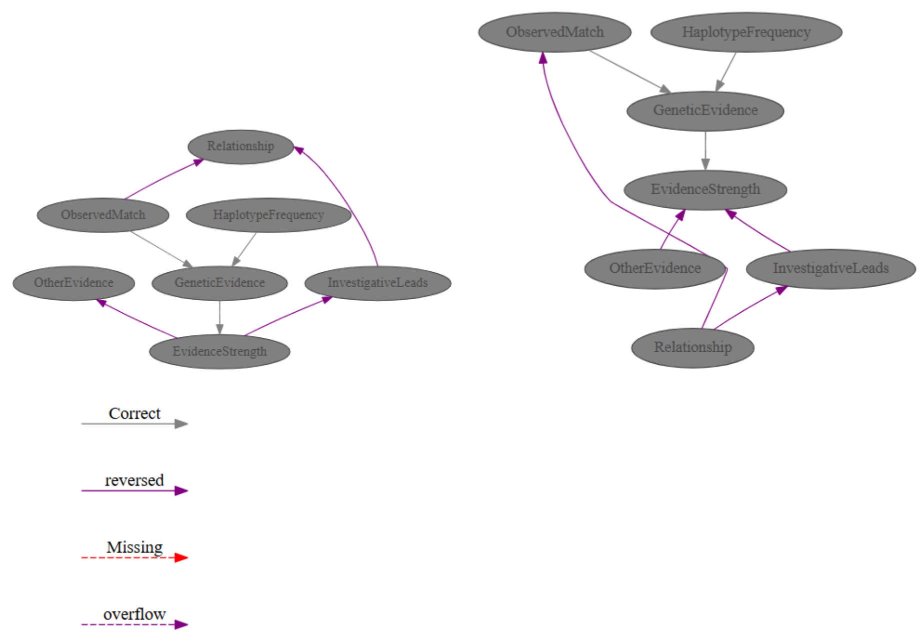

Figure 19: MIIC-algorithm, learned structure from 500 samples.

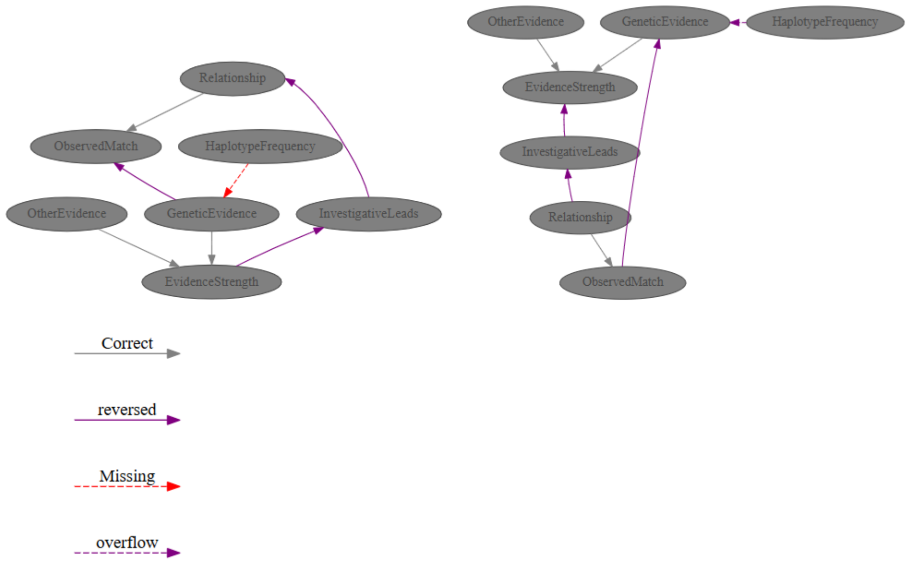

Figure 20: MIIC-algorithm, learned structure from 1000 samples.

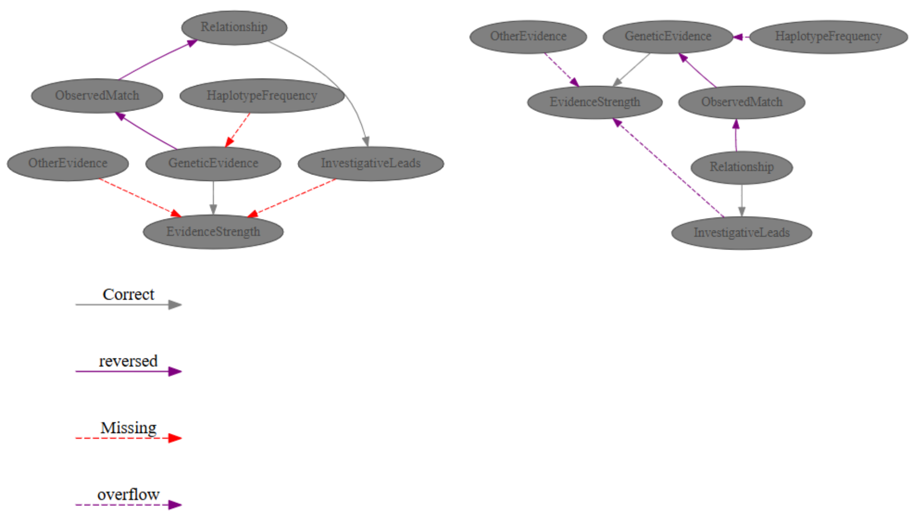

Figure 21: GHC-algorithm, learned structure from 100 samples.

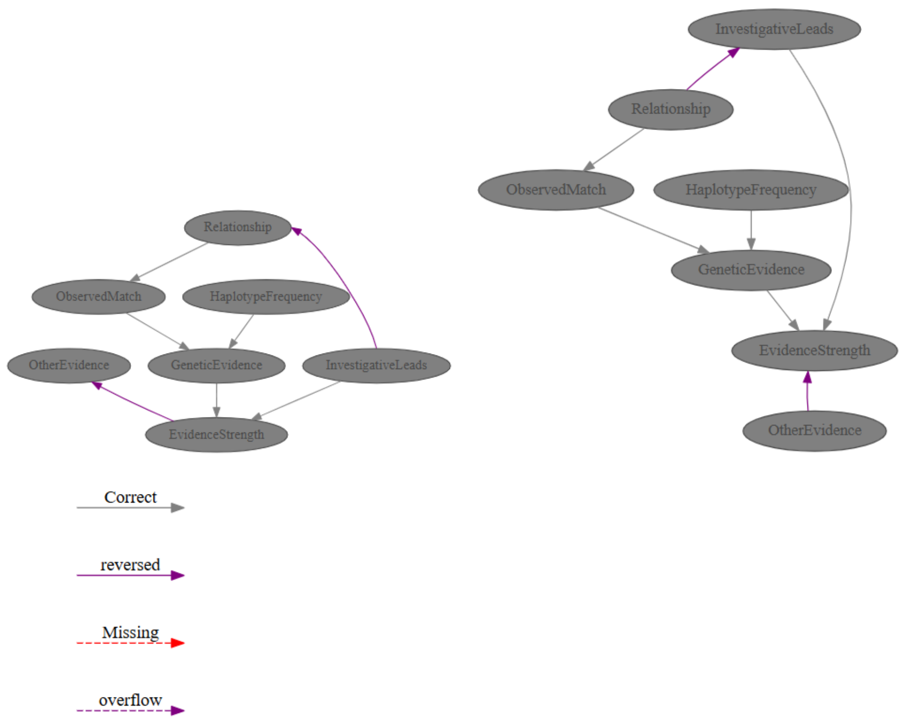

Figure 22: GHC-algorithm, learned structure from 500 samples.

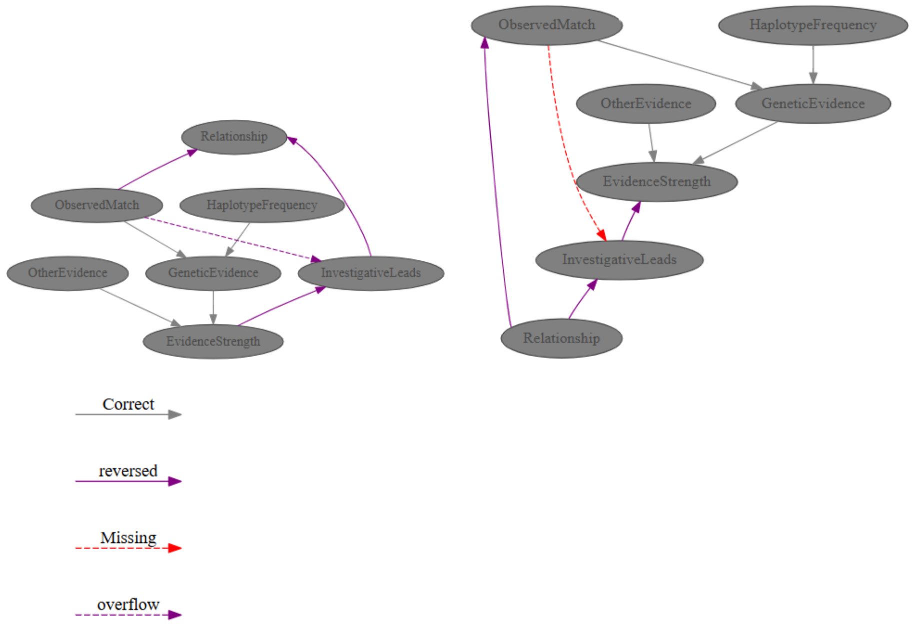

Figure 23: GHC-algorithm, learned structure from 1000 samples.